	Composition		Networks			Statistical information				Numerical conditioning	
Scenario	Shannon $H^\star$	$\|\boldsymbol{p}\|_2^2$	Edge density	Mean degree	Hoyer sparsity	Mean overlap $\bar{\omega}$	Max overlap $\omega_{\max}$	Hellinger	max f_{cov}	$\kappa\{\boldsymbol{\Sigma}(\boldsymbol{p})\}$	$\lambda_{\min}\{\boldsymbol{\Sigma}(\boldsymbol{p})\}$
balanced / low overlap											
CT1 Scale-free / CT2 Scale-free	1	0.333	0.1	1.9	0.329	0.0203	0.0662	0.971	0.387	5.49	0.463
CT1 SBM / CT2 Scale-free	1	0.333	0.0912	1.73	0.32	0.0208	0.064	0.971	0.463	3.77	0.491
CT1 Scale-free / CT2 SBM	1	0.333	0.0982	1.87	0.304	0.0206	0.059	0.971	0.442	3.18	0.544
CT1 SBM / CT2 SBM	1	0.333	0.0895	1.7	0.376	0.0195	0.063	0.971	0.412	3.56	0.495
balanced / moderate overlap											
CT1 Scale-free / CT2 Scale-free	1	0.333	0.1	1.9	0.329	0.0765	0.194	0.907	0.718	5.49	1.86
CT1 SBM / CT2 Scale-free	1	0.333	0.0912	1.73	0.32	0.0802	0.202	0.897	0.833	3.77	2.84
CT1 Scale-free / CT2 SBM	1	0.333	0.0982	1.87	0.304	0.0817	0.209	0.895	0.812	3.18	2.96
CT1 SBM / CT2 SBM	1	0.333	0.0895	1.7	0.376	0.0802	0.199	0.9	0.756	3.56	2.18
balanced / high overlap											
CT1 Scale-free / CT2 Scale-free	1	0.333	0.1	1.9	0.329	0.2	0.293	0.769	0.926	5.49	9.19
CT1 SBM / CT2 Scale-free	1	0.333	0.0912	1.73	0.32	0.194	0.248	0.764	0.968	3.77	17.4
CT1 Scale-free / CT2 SBM	1	0.333	0.0982	1.87	0.304	0.198	0.266	0.769	0.957	3.18	15.4
CT1 SBM / CT2 SBM	1	0.333	0.0895	1.7	0.376	0.199	0.28	0.764	0.944	3.56	11.8
moderately unbalanced / low overlap											
CT1 Scale-free / CT2 Scale-free	0.5	0.719	0.1	1.9	0.355	0.0354	0.114	0.971	0.71	9.48	0.769
CT1 SBM / CT2 Scale-free	0.5	0.719	0.0912	1.73	0.606	0.041	0.126	0.971	0.794	7.39	0.887
CT1 Scale-free / CT2 SBM	0.5	0.719	0.0982	1.87	0.354	0.0369	0.113	0.971	0.764	9.13	0.811
CT1 SBM / CT2 SBM	0.5	0.719	0.0895	1.7	0.608	0.0406	0.113	0.971	0.78	7.39	0.911
moderately unbalanced / moderate overlap											
CT1 Scale-free / CT2 Scale-free	0.5	0.719	0.1	1.9	0.355	0.132	0.362	0.907	0.908	9.48	3.1
CT1 SBM / CT2 Scale-free	0.5	0.719	0.0912	1.73	0.606	0.157	0.404	0.897	0.957	7.39	5.13
CT1 Scale-free / CT2 SBM	0.5	0.719	0.0982	1.87	0.354	0.143	0.365	0.895	0.946	9.13	4.42
CT1 SBM / CT2 SBM	0.5	0.719	0.0895	1.7	0.608	0.151	0.403	0.9	0.94	7.39	4.02
moderately unbalanced / high overlap											
CT1 Scale-free / CT2 Scale-free	0.5	0.719	0.1	1.9	0.355	0.316	0.504	0.769	0.98	9.48	15.3
CT1 SBM / CT2 Scale-free	0.5	0.719	0.0912	1.73	0.606	0.341	0.514	0.764	0.993	7.39	31.5
CT1 Scale-free / CT2 SBM	0.5	0.719	0.0982	1.87	0.354	0.301	0.446	0.769	0.989	9.13	23
CT1 SBM / CT2 SBM	0.5	0.719	0.0895	1.7	0.608	0.336	0.559	0.764	0.988	7.39	21.8
highly unbalanced / low overlap											
CT1 Scale-free / CT2 Scale-free	0.1	0.961	0.1	1.9	0.355	0.0756	0.235	0.971	0.755	9.78	1.01
CT1 SBM / CT2 Scale-free	0.1	0.961	0.0912	1.73	0.613	0.1	0.302	0.971	0.824	7.58	1.17
CT1 Scale-free / CT2 SBM	0.1	0.961	0.0982	1.87	0.355	0.0836	0.257	0.971	0.81	9.77	1.03
CT1 SBM / CT2 SBM	0.1	0.961	0.0895	1.7	0.613	0.095	0.28	0.971	0.812	7.58	1.2
highly unbalanced / moderate overlap											
CT1 Scale-free / CT2 Scale-free	0.1	0.961	0.1	1.9	0.355	0.221	0.589	0.907	0.926	9.78	4.06
CT1 SBM / CT2 Scale-free	0.1	0.961	0.0912	1.73	0.613	0.288	0.703	0.897	0.964	7.58	6.78
CT1 Scale-free / CT2 SBM	0.1	0.961	0.0982	1.87	0.355	0.238	0.605	0.895	0.959	9.77	5.6
CT1 SBM / CT2 SBM	0.1	0.961	0.0895	1.7	0.613	0.271	0.714	0.9	0.95	7.58	5.3
highly unbalanced / high overlap											
CT1 Scale-free / CT2 Scale-free	0.1	0.961	0.1	1.9	0.355	0.477	0.73	0.769	0.984	9.78	20
CT1 SBM / CT2 Scale-free	0.1	0.961	0.0912	1.73	0.613	0.544	0.778	0.764	0.994	7.58	41.6
CT1 Scale-free / CT2 SBM	0.1	0.961	0.0982	1.87	0.355	0.448	0.668	0.769	0.992	9.77	29.1
CT1 SBM / CT2 SBM	0.1	0.961	0.0895	1.7	0.613	0.52	0.823	0.764	0.99	7.58	28.7

Mean signature, the same in every scenario: $\kappa(\boldsymbol{\mu}) = 4.38$, $\lambda_{\min}(\boldsymbol{\mu}^\top \boldsymbol{\mu}) = 10$.
Green: easiest value of a column; red: hardest. Easiest is the highest number of cells, Shannon entropy, Hoyer sparsity, Hellinger distance, covariance share and $\lambda_{\min}$; the lowest cell-size ratio, concentration, edge density, mean degree, overlap and condition number. Shading and size: Shannon entropy, mean overlap and condition number.